

To appear in C. Howe, P. Chamorro, T. Gupton, and M. Renwick (Eds.), *Theory, data, and practice. Selected papers from the 49th Linguistic Symposium on Romance Language*, Open Romance Linguistics 1, pp. 85–106. Berlin : Language Science Press.

On the pragmasemantics of a high adjunct wh-word

Damien Fleury

University of Paris

Lucia M. Tovenà

Université de Paris

Questions with *comment* (how) in French may allow a reason reading when they occur in sentence initial position, besides manner and means reading. The paper provides empirical evidence in support of an analysis of this type of *comment* as externally merged in the left periphery of the clause, adapting existing proposals for reason ‘why’ counterparts. These questions convey the information that the proposition about the described situation, hereafter the prejacent, disconfirms the speaker’s expectations. They express her attempt to get information to resolve this conflict. *Comment* signals this particular function of the question and works as a discourse management device that aims to align the viewpoints of the interlocutors about preconditions to admitting the prejacent into the common ground.

1 Issues about reason-*comment* (how)

1.1 When *comment* (how) is asking about reasons

The ‘canonical’ interpretation of *comment* (how) in French is as a wh-item that allows one to question about manners and means, see (1a) and (1b) respectively.¹

¹The abbreviations used in this paper are glossed as follows: Cl: clitic personal pronoun; COND: conditional mood; DAT: dative case; Neg: negation marker; T: epenthetic ‘t’ consonant.

Moreover, questions with *comment* such as (2), allow a reason reading, highlighted by the possible congruent answer (2b). Possible answers with the manner and means readings are still available, but less salient in this context.²

- (1) Comment Daniel tourne-t-il cette plaque ?
how Daniel turn-T-he this plate
How does Daniel turn this plate?
a. Lentement (manner)
Slowly
b. Avec un tournevis (means)
With a screwdriver.
- (2) Q: Comment Max peut-il lire le courrier de Paul ?
how Max can-he read the mail of Paul
How can Max read Paul's mail?
a. A: Il est curieux (reason)
He is nosy
b. A: A la hâte
Cursorily
c. A: Avec une loupe
With a hand magnifier

Question (2), under its reason reading, has the form of a wh question, but is not a 'standard' constituent question at least insofar as the wh-item is understood not to bind a variable within the IP syntactic structure. Neither is it a yes/no question, for that matter, and cannot receive a yes/no answer. The question is about the reasons the event of 'Max reading Paul's mail' could come about. It is an inquiry about elements of (possibly private) explanatory theories that would support some data. Such data, which can be contextually available, are recalled in the question by the proposition expressed by the clause that would correspond to the sentence radical in the manner and reason readings were it not for the

² According to Cysouw (2004), the interrogative category of 'manner' is only lexicalised in about 40% of the world's languages. In many of these, wh-words or expressions with this meaning can form questions with reason flavoured readings. These questions differ in their pragmatic content, and a characteristics of reason-*comment* questions is that they convey the information that the prejacent disconfirms the speaker's expectations. Reason readings are available when the wh-words occur in the highest layer of the structure of the sentence, a.o. see (Rizzi 2001; Ko 2005; Tsai 2008). How these readings relate to manner and what is their affinity with modality are research issues beyond the scope of this paper.

absence of a gap in the IP. Here, the sentence radical is saturated, if we can say so when comparing this questions with others about adjuncts, and is interpreted as a proposition, not as a propositional function. Such a proposition is referred to as ‘prejacent’ hereafter. The prejacent in sentence (2) is the proposition ‘Max lit le courrier de Paul’.

Another specificity of questions like (2) is that they show some characteristics usually associated with exclamation and mirativity, though the intonation is not typical of exclamatives. Namely, they typically convey the information that the situation described by the prejacent disconfirms the speaker’s expectations. At times, they may carry a hint of disapproval, or have a rhetorical flavour.

1.2 Some strands of analysis

The reason reading of *comment* in question (2) concerns the reasons the situation of ‘Max reading Paul’s mail’ could come about, as just noted. This reading makes *comment* somewhat close to ‘why’, and is reminiscent of the why-how alternation discussed by (Collins 1991; Ochi 2004; Tsai 2008) among others. It is worth underscoring that the cases of why-how alternation discussed in the literature seem to relate to the reason readings of these two wh-items, not to the wh-items in themselves. As an aside, it should also be noted that *comment* questions differ from these cases of alternation because the prejacent describes a situation that is not necessarily true, as it may be potential or actual. Therefore, it is not presupposed, although this might not appear from the English translations. Factors enhancing the factuality of the prejacent are e.g. the use of the passé composé in the clause, as illustrated by (3). Factors going against it are e.g. the use of a conditional verb form, as illustrated by (4). In all cases, the prejacent conveys at issue information, and the reason(s) for the (potential) actualisation of the situation it describes are focussed on in the question.

- (3) Comment Max a lu le courrier de Paul ?
how Max has read the mail of Paul
How comes that Max read Paul’s mail?
- (4) Comment Max aurait lu le courrier de Paul ?
how Max have.COND read the mail of Paul
How could it be that Max read Paul’s mail?

Reason questions can also be worded with *pourquoi* (why) in French, see (5). Indeed, *pourquoi* can be replaced by *comment* in cases such as (5), when the question is used to inquire about the conditions that led to formulating the request for a deadline.

- (5) Pourquoi il doit te répondre avant mardi ?
 why he must you-DAT answer before Tuesday
 Why should he answer to you by Tuesday?

However, *pourquoi* does not alternate with *comment* when used for inquiring about the goal/purpose of the initiator of the event, or the situation that would ensue (result). The example in (6a) presents a wh-question that is acceptable and has a reason interpretation with either wh-word. However, only the version with *pourquoi* is compatible with the continuation given in (6b), as indicated with the diacritic ‘#/ok’.

- (6) a. Comment/pourquoi l’agent de police tolère-t-il un tel désordre ?
 how/why the-police-officer allow-T-he a such mess
 How come/why the police officer allows such a mess?
 b. #/ok Pour éviter une confrontation inutile
 to avoid a confrontation unnecessary
 In order to avoid unnecessary confrontation.

As we said, *comment* interpreted as inquiring about a reason occurs in sentence initial position, where it is understood not to bind a variable low in the syntactic structure, below the IP node. The semantic intuition that *comment* works as operating over the rest of the sentence from outside the clause, questioning some conditions about the prejacent *p*, can be substantiated by extending to this case syntactic proposals put forth for reason *why*. In the literature, it has been argued that the adjunct *why* in questions like *Why did Max leave?* is externally merged in the left periphery of the clause and does not bind any syntactic variable inside the IP, cf. (Rizzi 1990; 2001), (Shlonsky & Soare 2011) among others. A similar analysis may be explored for *comment* with a reason reading in (2). In section 2, we provide empirical support for it. An expected advantage is that the reason reading of the two wh-expressions *comment* and *why* would receive a similar characterisation that is compatible with the discourse role of the prejacent.

Next, as we mentioned above, reason-*comment* questions show some characteristics usually associated with exclamation and mirativity. Surprise can be conveyed by other linguistic expressions in French, whose use, in turn, is not limited to the expression of surprise, e.g. wh-exclamatives (7).

- (7) Qu’est-ce qu’il est grand !
 what-is-it that-he is tall
 How tall is he!

The characterisation of reason-*comment* that we suggest partially overlaps analyses of exclamatives and mirativity, insofar as we assume that reason-*comment* operates on speaker's expectations. We assume that in asking a *comment* question with a reason reading, the speaker initiates an attributional search in reaction to an expectation disconfirmation, following Fleury & Tovenia (2018). In social psychology, attributions (Heider 1958) are the explanations humans come up with in order to understand the causes of behaviours, actions and events. The search verbalised with a *comment* question aims at finding some proposition(s) to which to attribute the resolution of the conflict between the situation described by the prejacent and the speaker's expectation.

These questions are interrogative utterances where *comment* seems to function as a discourse management device, adapting an idea from (Krifka 2008). They do not seek factual information to be added to the common ground. Rather, they serve to negotiate with the interlocutors about restrictions on the future continuations of the conversation. The speaker puts on hold a (possibly implicit) invitation from the interlocutors to admit the prejacent into the common ground and tackles preconditions to admitting it, which is represented by sorting out the contradiction between the prejacent and her expectations.

The speaker uses a question with reason-*comment* in her attempt to get information for recomputing her expectations. The truth value of the prejacent is not fixed, as it may characterise actual or potential situations in the context, and the question does not seek information for fixing it. We characterise the speaker's expectations—called the *expectation set*—as a minimal set of propositions that makes the prejacent false, see section 3. This expectation set partitions the possible worlds. Our proposal builds on work by (Lewis 1988) and (Von Stechow & Gillies 2010), where partitions on possible worlds are used to define some relation of relevance between propositions (Lewis 1988) or to represent speaker's direct knowledge (Von Stechow & Gillies 2010). The relations between the prejacent, the propositions in the expectation set and the answer, are addressed by means of the partition of the possible worlds triggered by the expectation set.

Plausible attributions count as congruent answers. Their relevance is captured in terms of operations that reason-*comment* can do with them on the expectation set, as discussed in subsection 4.1. Subsection 4.2 offers a detailed discussion of an example, with the aim to show how relevant information can operate on the partition corresponding to the speaker's expectation and lead to an acceptable answer. Section 5 concludes the paper.

2 High adjunct wh-word

The suggestion that reason *comment* questions such as (2) have a meta-conversational function gets support from facts about the syntactic analysis that can be proposed for these questions in this specific reading. If these are questions asking about information in the common ground or on participants' momentary belief sets, we expect the wh-item to pertain to the portion of discourse-related syntactic projections in the left periphery of a sentence. In this section, we provide a number of empirical observations that motivate the adoption of an analysis for reason-*comment* as merged high.

First, reason-*comment* typically occurs in clause initial position and not in a lower putative in-situ position. The congruent answers highlight the unavailability of this reading for the question in (8). Embedded cases appear to be limited to occurrences under intensional question predicates (9a), not under extensional question predicates (9b). The complement clause of verbs like *savoir* are not unambiguously interpreted as questions. This piece of evidence suggests that reason-*comment* is a question word in its reason reading. More research is needed to find out if it is a root phenomenon.

- (8) Tu lui as répondu comment ?
 you she.DAT have answered how
 How did you answer to her?
- a. Calmement. (manner)
 Calmly.
- b. Par courrier. (means)
 By mail
- c. # J'étais obligé, c'est mon patron. (reason)
 I had to, he's my boss.
- (9) a. Les enfants se demandent comment ça n'arrive pas dans le
 the children Cl ask how that Neg-arrive Neg in the
 monde réel.
 world real
 The children wonder why that does not happen in real world.
- b. #Les enfants savent comment Max a lu le courrier de Paul
 the children know how Max has read the mail of Paul
 The children know how Max read Paul's mail.

Second, reason-*comment* is not in the scope of propositional negation contained in the clause where it would be construed and that expresses the preja-

cent. The reason interpretation is the only one available in (10), see the answers. This piece of evidence provides information on how low reason-*comment* could be merged. The contrast tells us that it must be at least above NegP.³

- (10) Comment Max n'a-t-il pas lu le courrier de Paul ?
 how Max Neg-has-T-he Neg read the mail of Paul
 How could Max not have read Paul's mail?
- a. Il n'est pas curieux.
 He isn't nosy
 - b. # (Pas) A la hâte manner
 - c. # (Pas) Avec une loupe means

Third, reason-*comment* does not scopally interact with quantifiers in the clause. Sentence (11) is interpreted with *comment* outscoping the quantifier *tout le monde* (everyone), matching the linear scope (11a). The inverse scope reading where *tout le monde* outscopes reason-*comment* would rely on a *wh*-trace inside the IP and is not available (11b). The sentence does not have functional and pair-list readings. Notice that the quantifier can outscope manner *comment* (12).

- (11) Comment tout le monde déteste Daniel ?
 how all-the-world hate Daniel
 How come everyone hates Daniel?
- a. What is the reason *q* such that the fact that everybody hates Daniel is due to reason *q*?
 - b. # What is the reason *q* such that for each *x*, *x* hates Daniel for the reason *q*?
- (12) a. Comment tout le monde parle à Daniel ?
 how all-the-world talk to Daniel
 How does everyone talk to Daniel?
- b. What is the manner *m* such that for each *x*, *x* talk to Daniel in manner *m*?

Sentence (13) may have a pair-list reading with manner-*comment*, see the possible answer in (13a) and the interpretation of the question in (13b).

³The unavailability of the manner reading would follow from an intervention effect (Rizzi 1990) or from not meeting D-linking due to the fact that there isn't a meaningful unique maximal entity in the negative extension of the interrogative expression without contextual restriction (Rullmann 1995).

- (13) Comment les invités continuent-ils à s’amuser ?
 how the guests continue-they to have-fun
 How/how come the guests continue to have fun?
- a. Léa danse et Paul chante.
 Léa is dancing and Paul is singing.
 - b. For each guest x, what is the manner m such that x continues to have fun in manner m?

Now imagine a context where one of the guests has just been taken to hospital. The speaker may expect that the guests stop having fun. Then, sentence (13) may be interpreted as a reason-*comment* question with the wh-item taking wide scope, as in (14b), as confirmed by the acceptability of the continuation in (14a) in such a context.

- (14) a. Ils sont inconscients.
 They are thoughtless.
- b. What is the reason q such as the fact that the guests continue to have fun is due to reason q?

However, the pair-list reading is not allowed for the reason reading, as shown by the unacceptability of an answer such as in (15a) and the interpretation of the question in (15b).

- (15) a. # Léa est inconsciente et Paul est fou.
 Léa is thoughtless and Paul is mad.
- b. # For each guest x, what is the reason q such that the fact that x continue to have fun is due to reason q?

Finally, wh-questions license negative polarity items, but the wh-word *why* does not, and neither do questions with reason-*comment*. The judgements on the naturalness of sentences like (16a) vary according to whether the speakers use the expression *quoi que ce soit* (anything) productively, but all speakers agree on the acceptability contrast between (16a) and (16b).

- (16) a. # Comment Daniel a dit quoi que ce soit d’intéressant ?
 how Daniel has said anything of-interesting
 How come Daniel said anything interesting?
- b. A qui/Quand Daniel a dit quoi que ce soit d’intéressant ?
 to whom/when Daniel has said anything of-interesting
 To whom/When did Daniel say anything interesting?

The evidence marshalled in this section—mostly based on weak island and scoping effects—strongly supports the hypothesis according to which *comment* with a reason reading must be generated high. For completeness, we observe that the reason reading is often facilitated by other factors. The reason reading is favoured for instance when the preajcent is syntactically realised as a separate clause, e.g. a clausal complement of attitude or opinion verbs (17), and when the preajcent is embedded under a modal verb or negation, cf. (18)–(19).

- (17) Comment Max ose lire le courrier de Paul ?
how Max dare read the mail of Paul
How does he dare to read Paul's mail?
- (18) Comment Max peut-il lire le courrier de Paul ?
how Max can-he read the mail of Paul
How can he read Paul's mail?
- (19) Comment Max ne lit pas le courrier de Paul ?
how Max Neg read Neg the mail of Paul
How could it be that he doesn't read Paul's mail?

Some of these factors, as well as the presence of semi-modals and high degree expressions are known to favour rhetorical readings of questions (Sadock 1971; Desmets & Gautier 2009). The phenomenon is illustrated in (20). More research is needed on this topic.

- (20) a. Comment Léa ose lire le courrier de Tom ?
how Lea dare read the mail of Tom
How does Léa dare to read Tom's email?
- b. Comment peux-tu quitter un homme aussi adorable ?
how can-you leave a man so lovely
How can you break up with such a sweet man?

Summing up, reason *comment*-questions were suggested to have a meta-conversational function in section 1, as confirmed by the high positioning of reason-*comment* in the syntactic structure defended in this section 2. Indeed, the high positioning goes together with the interpretation of the IP as a full proposition, i.e. the preajcent, and the fact that this proposition is to be examined in relation to the speaker's expectations. Expectations are a form of knowledge/information the speaker relies on in her understanding and reasoning about the world. Our proposal is going to dovetail the use of a partition of possible worlds to represent the speaker's knowledge with its use to represent the possible answers to the

question posed by the speaker. To this end, we address the topic of expectations in the next section.

3 Expectation set and the notion of issue

3.1 Expectations and answers

Before we see how *reason-comment* operates on the speaker's expectations, we have to characterise these expectations, and determine the status of the prejacent and the possible answers with respect to these expectations.

Questions are analysed as sets of alternatives, following Groenendijk & Stokhof (1984). Standardly, yes/no questions like *Max lit-il le courrier de Paul ?* (Does Max read Paul's mail?) are assumed to partition the context set into p -worlds and $\neg p$ -worlds. Wh-questions like *Who read Paul's mail?* denote sets of possible contextually defined alternatives, where the values for the wh-element fill the gap in the sentence radical.

Intuitively, *reason-comment* questions do not seem to fit straightforwardly into this classification, because they do not give rise to any alternative *per se*, computed using the sentence radical. The alternatives depend on the current expectations of the speaker and on the information that she can get from the context. More precisely, they depend on the way the speaker may modify her current expectations given some new information. Furthermore, an answer to a *reason-comment* question may be non exhaustive but sufficient to provide enough information to the speaker.

In a question with *reason-comment*, the situation described by the prejacent p disconfirms the speaker's expectations, but only some expectation(s) make(s) the prejacent false. We assume a contextually available set of speaker's personal expectations, containing only propositions relevant for the truth of the prejacent at the time the question with *comment* is uttered. We note *Exp* this *expectation set*.

In the example (21), the speaker's expectation may contain specific information, e.g. $p_1 = \text{Max is respectful}$, and a causal schema, e.g. $p_1 \rightarrow \neg p = \text{if Max is respectful then he doesn't read the mail of others}$ (and specifically Paul's), thus we can assume $\text{Exp} = \{p_1, p_1 \rightarrow \neg p\}$. The speaker expects propositions p_1 and $p_1 \rightarrow \neg p$ to be true and to characterise the actual world where (21a) is uttered.

- (21) a. Comment Max lit le courrier de Paul ?
 how Max read the mail of Paul
 How (is it/could it be) that Max read Paul's mail?

- b. Il est curieux
He is nosy

The propositions of *Exp* make the prejacent *non contingent* in the worlds faithful to the speaker's expectations, more specifically false in all these worlds, that is $(\bigcap \text{Exp}) \cap p = \emptyset$.⁴ Recall that a proposition is non-contingent with respect to a set of propositions if it is true in all the worlds faithful to this set of propositions, or false in all these worlds. In our example (21), the prejacent *Max lit le courrier de Paul* (p) is not compatible with the expectations of the speaker, p_1 and $p_1 \rightarrow \neg p$.

In asking a question with reason-*comment*, the speaker initiates a search in reaction to an expectation disconfirmation, aiming at finding some proposition that may resolve the conflict between the situation described by p and the speaker's expectations *Exp*, and that makes p *contingent*, i.e. there is at least one p -world in the intersection of the new expectation set *Exp'*: $(\bigcap \text{Exp}') \cap p \neq \emptyset$. The p -worlds are represented by the right grey circle in Figure 1, the worlds faithful to the speaker's initial expectations, $\bigcap \text{Exp}$, by the left circle drawn with a solid line, and those faithful to the speaker's new expectations, $\bigcap \text{Exp}'$, by the dotted line circle.

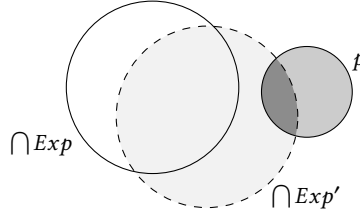


Figure 1: Intersection with p

The answer *il est curieux* (he is nosy) in (21b) may be considered relevant by the speaker, regardless of whether she considers it to be true, possible or false. Informally, the answer *il est curieux* (q) could lead the speaker to have new expectations *Exp'*, for instance that Max may actually not be always respectful and that curiosity may lead Max to read Paul's mail. Even if the speaker does not accept the truth or the possibility that Max is nosy (q) she may nevertheless consider this proposition relevant and explore what would be the case if the answer were

⁴ $(\bigcap \text{Exp}) \cap p$ is the intersection of the worlds faithful to the speaker's expectations ($\bigcap \text{Exp}$) with the p -worlds. The intersection is empty, i.e. $(\bigcap \text{Exp}) \cap p = \emptyset$, iff p is false in all the worlds faithful to the speaker's expectations, i.e. the proposition p is incompatible with these expectations.

true. The answer q , if considered relevant, will be a congruent answer if it makes the prejacent contingent in the new expectation set Exp' , that is $(\bigcap \text{Exp}') \cap p \neq \emptyset$.

Moreover, we will see that the prejacent p cannot be reduced to a logical combination of the propositions in Exp in non-trivial cases,⁵ whereas the new information q sought by the speaker may or may not be such a combination. This kind of combinations—called *issues* following (Von Fintel & Gillies 2010), and discussed in details in section 3.3—are relevant in that they allow us to identify different ways of recomputing the expectation set. Before discussing that, we consider the minimality of the expectation set, in section 3.2.

3.2 A minimal set of expectations

Exp is a *minimal* set of propositions making the prejacent false, whenever removing a proposition from it, say p_1 , is enough to make the prejacent p contingent with respect to it. More precisely, the prejacent is contingent with respect to the new expectation set Exp^- obtained from Exp by removing p_1 , that is $\text{Exp} - \{p_1\}$. The p -worlds are represented by the dark grey area in Figure 2, the intersection of Exp by the area inside the smaller circle, and the intersection of Exp^- by the area inside the bigger circle. The intersection of Exp does not intersect the p -worlds, while the intersection of Exp^- does. Removing p_1 from Exp allows the speaker to conceive more worlds as possible, notably some world(s) where the prejacent is true.

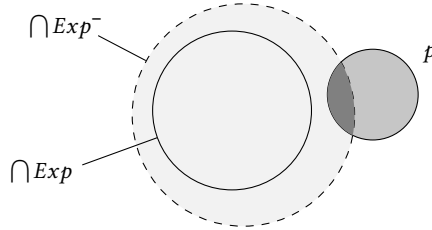


Figure 2: Minimality of Exp

Moreover, the speaker may entertain different expectations about the truth of the prejacent, which lead to different expectation sets. For instance, imagine a situation where John leaves a meeting before it is finished. The speaker may consider this behaviour unacceptable on the basis of the rules. But she also may

⁵Such a combination is obtained by using the logical operators $\neg$, $\wedge$, $\vee$. If we have $\text{Exp} = \{p_1, p_2\}$, then the propositions p_1 , $\neg p_1$, $p_1 \wedge p_2$ and $\neg p_1 \vee \neg p_2$ are some examples of this kind of combination.

consider it unusual regarding John's habits. These two judgements correspond to two expectation sets, one about rules (Exp in Figure 3) and one about habits (Exp_1).

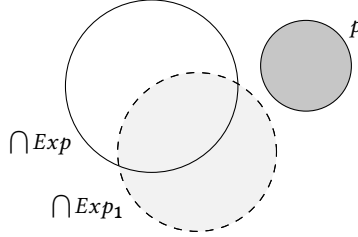


Figure 3: Two expectation sets

Neither the intersection of Exp nor that of Exp_1 intersect p in Figure 3. When the speaker asks the question *comment peut-il quitter la réunion avant la fin ?* (how can he leave the meeting before the end?), she has some expectations incompatible with the prejacent p . But she may have different judgements corresponding to different expectations sets at the same time, that are incompatible with the prejacent p . What is crucial is our assumption that a question is associated with one and only one contextually relevant expectation set.

The propositions of Exp partition the possible worlds into separate cells in a standard way.⁶ $\cap Exp$ is clearly one of the cells of the partition, namely the set of worlds where all the propositions in Exp are true.

Now we want to define more precisely this partition and look at the information available to the speaker (her expectations, the prejacent, the new information) in terms of that partition. We are going to present the link between contingency and partition, and sketch the forms of modification the new information leads to.

3.3 Subject matter and issues

The notion of contingency is modelled by the non equivalence among possible worlds, so we need a way to compute equivalence classes among worlds. Two worlds are considered alike if the relevant propositions have the same truth values on these two worlds (Lewis 1988). This being alike is an equivalence relation that Lewis called *subject matter*. The equivalence classes of such a relation are the cells of a partition on the possible worlds W . Each of its cells is a maximally

⁶For each cell, the propositions in Exp have the same truth values in all the worlds of the cell, the cells do not overlap, and cover all the possible worlds.

specific way things might be with respect to relevant propositions. Intuitively, the minimal collection of relevant expectations can be seen as a thematic way to select worlds as a function of the expectations they are consistent with.

In example (21), repeated in (22), the relevant propositions are *Max is respectful* (p_1) and *if Max is respectful then he doesn't read the mail of others* (and specifically Paul's) ($p_1 \rightarrow \neg p$).

- (22) a. Comment Max lit le courrier de Paul ?
 How (is it/could it be) that Max read Paul's mail?
 b. Il est curieux (He is nosy)
 c. Max n'est pas respectueux (Max is not respectful)

Assume S_{Exp} is the subject matter that corresponds to the propositions in the expectation set $Exp = \{p_1, p_1 \rightarrow \neg p\}$. The proposition p_1 is represented by the hatched area in Figure 4, and the proposition ($p_1 \rightarrow \neg p$) by the grey area.⁷ The cells of the partition induced by these two propositions (i.e. the equivalence classes of S_{Exp}) are delimited by thick lines. The partition contains the cells $p \wedge p_1$ (white hatched area), $\neg p_1$ (non-hatched grey area) and $\neg p \wedge p_1$ (grey hatched area), i.e. all the combinations of the two relevant propositions p_1 and ($p_1 \rightarrow \neg p$), by using the logical connectives $\neg$ and $\wedge$. There are only three cells in this example, because $\neg p_1 \wedge \neg(p_1 \rightarrow \neg p)$ is empty for logical reasons.⁸

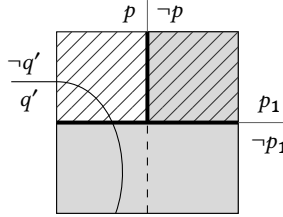


Figure 4: $S_{Exp}[q] \neq S_{Exp}$

A proposition q is considered an *issue* in the subject matter S_{Exp} iff $S_{Exp}[q] = S_{Exp}$, where $S_{Exp}[q]$ is the subject matter based on the proposition of Exp and q (Von Fintel & Gillies 2010). In plain words, a proposition is an issue in a subject matter when its informative contribution is already taken into account in the computation of the partition.

⁷By definition, $a \rightarrow b$ is equivalent to $\neg a \vee b$. Then the proposition $p_1 \rightarrow \neg p$ is equivalent to $\neg p_1 \vee \neg p$ and also to $\neg(p_1 \wedge p)$. It corresponds to the grey area in Figure 4.

⁸The intersection of $\neg p_1$ (non-hatched grey area in Figure 4) and $\neg(p_1 \rightarrow \neg p)$ (white hatched area) is empty.

Issues in S_{Exp} are non-contingent on each equivalence class of the subject matter S_{Exp} . In particular, p_1 and $p_1 \rightarrow \neg p$ are issues in S_{Exp} .⁹ By contrast, the proposition q' , as represented by a curved line in Figure 4, is not an issue in the subject matter S_{Exp} , for the cell $p \wedge p_1$ contains both q' -worlds and $\neg q'$ -worlds, and the same applies to the cell $\neg p_1$. In other words, the proposition q' is contingent on the cells $p \wedge p_1$ and $\neg p_1$, and we have $S_{Exp}[q'] \neq S_{Exp}$, more precisely: $S_{Exp}[q'] = \{p \wedge p_1 \wedge q, p \wedge p_1 \wedge \neg q, \neg p_1 \wedge q, \neg p_1 \wedge \neg q, \neg p \wedge p_1\}$.

We will say that the expectation set is *trivial* with respect to the prejacent p if p is a combination of the propositions in the expectation set. In order to clarify the notion of combination that we use, let us give an example with three propositions p_1 , p_2 and p_3 . The proposition $(p_1 \wedge p_2 \wedge \neg p_3) \vee (p_1 \wedge \neg p_2 \wedge p_3) \vee (p_1 \wedge \neg p_2 \wedge \neg p_3)$ is one of the combinations of the propositions p_1 , p_2 and p_3 .

In example (23), the prejacent p is *Max va au cinema* (Max go to the cinema). Suppose the speaker thinks that a world where Max go to the cinema is not possible under any circumstances, and that there is no particular reason for that. In this case, the expectation set is simply $\{\neg p\}$ and p is clearly a combination of the only proposition in the expectation set. The expectation set is trivial with respect to the prejacent and the question in (23) is not very natural.

- (23) Comment Max peut-il aller au cinéma ?
how Max can-he go to-the cinema
How can Max go to the cinema?

Usually, what the speaker thinks about the truth or the possibility of p relies on some kind of reasoning about circumstances, rules, norms or regularities. For instance, Max is not used to going to the cinema or Max has to finish a book. In this case, the prejacent p is not just a simple combination of the propositions in the expectation set. Here, we suppose the expectation is not trivial with respect to the prejacent. Then the prejacent cannot be an issue in the subject matter S_{Exp} , for an issue in the subject matter S_{Exp} is a combination of propositions in Exp . In example (22), it can be easily seen that the prejacent is contingent in some equivalence classes ($\neg p_1$), but non-contingent in some others (true in $p \wedge p_1$ and false in $\neg p \wedge p_1$). So the prejacent p is not an issue in the subject matter S_{Exp} .

⁹Propositions p_1 and $p_1 \rightarrow \neg p$ are not the only issues in the subject matter. For instance, $p \vee \neg p_1$ is also an issue. It is easy to verify that $p \vee \neg p_1$ is non-contingent on each cell of the partition: $p \vee \neg p_1$ is true on $p \wedge p_1$, true on $\neg p_1$, and false on $\neg p \wedge p_1$. Moreover, each proposition that corresponds to a cell is an issue, for it is true on the cell and false on the other cells (for the cells are non-overlapping). Likewise, the proposition that corresponds to the entire logical space is true on every cell and hence is an issue.

However, the expectations of the speaker correspond to the cell $\neg p \wedge p_1$ (the intersection of Exp) where the prejacent is non-contingently false. This is why the speaker is stuck, and by asking a reason question with *comment*, she aims at undoing the status of non-contingency of the prejacent in her expectations. Moreover, a relevant answer may or may not be an issue.

4 Solving the conflict

4.1 Congruent answers

We have seen that by asking a reason question with *comment*, the speaker aims at undoing the non-contingent status of the prejacent in her expectations. As mentioned in section 3, this type of question does not give rise to alternatives per se, contra canonical wh-questions. What is a congruent answer cannot be stated in a standard manner (von Stechow 1991). Yet, it is not the case that anything will do for an answer. In this section we explore a way to characterise good answers.

To simplify things a bit, we suppose that the speaker's knowledge and belief belong to the common ground, i.e. the propositions shared by her and the addressee. The expectation set Exp of the speaker may include knowledge, belief and other judgements compatible with her knowledge and belief. Typically, these compatible judgements could be default judgements. They correspond to default rules that the speaker holds because of partly known information (Veltman 1996; Crespo et al. 2018).¹⁰ Moreover, they are not necessarily compatible with the addressee's knowledge and beliefs. The speaker is looking for some new information q compatible with her knowledge and beliefs, such that recomputing the expectation set with q (Exp') makes the prejacent contingent, i.e. there exists a possible world compatible with Exp' where q and the prejacent p are true.

The expectation set Exp partitions the possible worlds and, as seen in section 3.3, the new piece of information q may or may not be an issue with respect to this partition, and thus leads to different possible modifications of the expectation set. The purpose of this partition is to distinguish different kinds of modifications of the expectation set, not to represent the answers of the question.

When the speaker regards the answer q as relevant, she modifies her expectations Exp according to this new information. More precisely, the proposition q is in a kind of conflict with at least one proposition in Exp, say p' (otherwise the

¹⁰Crespo et al. (2018, p.269) remind us that: 'Default rules are of crucial importance when some decision must be made in circumstances where the facts of the matter are only partly known. They affect an agent's expectations and as such help to narrow down the range of possibilities an agent has to take into account'.

speaker does not need to modify her expectations). If q is an issue in the partition generated by Exp , typically when q is the negation of p' , then the speaker may revise Exp by replacing p' by q in it. In Paul's mail example, if q is the information that Max is not respectful, that is $q = \neg p_1$, then the speaker may replace p_1 by $\neg p_1$ in her expectations. By contrast, if q is not an issue, then she may replace p' by a proposition compatible with some possible worlds that are both q and $\neg p'$, while not necessarily considering the proposition q as true. In Paul's mail example, if q is the information that Max is nosy, then the speaker may consider that this information clashes to some extent with her expectation that Max is respectful (p_1). Indeed, the more someone is nosy (q), the more he or she is likely to be disrespectful ($\neg p_1$). This is not an implication between the fact that Max is nosy (q) and the fact that Max is disrespectful ($\neg p_1$). Nevertheless, it is commonly accepted—under some general and shared principles—that curiosity may lead to disrespect, in some circumstances. As a result, the speaker may accept the possibility that some q -worlds are also $\neg p_1$ -worlds, without rejecting $\neg q$ -worlds. Let's note Exp' these new expectations. If the speaker accepts the truth or the possibility of the proposition q , and if p is contingent in the new expectations Exp' , i.e. $\bigcap \text{Exp}'$ intersects p , then the proposition q is a good answer. Moreover, if she considers the proposition q true (not only possible), such a proposition is added to the previous expectation set Exp' . The new expectation set, i.e. $\text{Exp}'' = \text{Exp}' \cup \{q\}$, intersects p as well. But if the speaker rejects the truth of q , then the proposition $\neg q$ is added to the expectation set Exp' , and the new expectation set $\text{Exp}''' = \text{Exp}' \cup \{\neg q\}$ does not intersect p . The prejacent remains non-contingent and false with respect to Exp''' . The proposition q is a bad answer.

The modifications the answer q may cause on the expectation set differ depending on whether the answer is an outcome or not. We look at this distinction with an in-depth discussion of an example in section 4.2.

4.2 Computing the expectation set

Relevant answers provide information that alters the non-contingent status of the prejacent, whether or not they are an issue in the subject matter. Recall that, roughly speaking, a proposition is an issue in the subject matter if its informative contribution is already taken into account.

4.2.1 Getting new information that is an issue

Assume question *Comment Max lit le courrier de Paul?* in (22a) gets the answer *Max is not respectful* (22c). This answer $\neg p_1$ is the negation of an expected proposition, and is an issue in the subject matter.

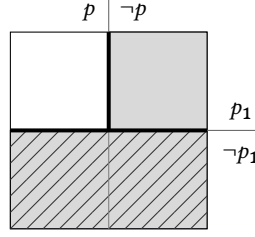


Figure 5: Exp'

If the speaker accepts answer $\neg p_1$, the subject matter is unchanged, but the new expectation set is $\text{Exp}' = \{\neg p_1, p_1 \rightarrow \neg p\}$ and the possible worlds are $\bigcap \text{Exp}' = \neg p_1$ (grey hatched lower half area in Figure 5). The proposition p becomes contingent in the expectation set Exp' and $\neg p_1$ is a good answer.

4.2.2 Getting new information that is not an issue

Consider question (22a), and the possible answer *Max est trop curieux* (Max is way too nosy) (q). Proposition q is not an issue in S_{Exp} , for it is contingent in at least one cell of the partition. Among the worlds where Max is respectful and does not read Paul's mail ($p_1 \wedge \neg p$), we can imagine some possible worlds where Max is nosy (but not to the point that he reads Paul's mail), and others where he is not nosy at all. In other words, the proposition q is contingent in the cell $p_1 \wedge \neg p$. If the speaker judges it relevant, it may be taken as a possible cause for the lack of respect by Max ($\neg p_1$), but does not entail it. In other worlds, among q -worlds (worlds where Max is nosy), p_1 -worlds as well as $\neg p_1$ -worlds seem possible to the speaker (Max may or may not be disrespectful when he is nosy). But $\neg q$ -worlds are still expected to be p_1 -worlds. So $\neg q \rightarrow p_1$ is expected. Moreover, the proposition $p_1 \rightarrow \neg p$ is still expected (if Max is respectful, he does not read Paul's mail). The new expectation set is $\text{Exp}'' = \{\neg q \rightarrow p_1, p_1 \rightarrow \neg p\}$, that is $(q \wedge \neg p_1) \vee (p_1 \wedge \neg p)$.¹¹

¹¹Visually, the intersection of $\neg q \rightarrow p_1$ (hatched area) and $p_1 \rightarrow \neg p$ (grey area) corresponds to the grey hatched area in Figure 6, that is the union of the top right square ($p_1 \wedge \neg p$) and the bottom half of q (i.e. $q \wedge \neg p_1$). We can also calculate it as follows :

$$\begin{aligned}
 & (\neg q \rightarrow p_1) \wedge (p_1 \rightarrow \neg p) \\
 &= (q \vee p_1) \wedge (\neg p_1 \vee \neg p) \\
 &= (q \wedge \neg p_1) \vee (q \wedge \neg p) \vee (p_1 \wedge \neg p_1) \vee (p_1 \wedge \neg p) \\
 &= (q \wedge \neg p_1) \vee (q \wedge \neg p) \vee (p_1 \wedge \neg p) \\
 &= (q \wedge \neg p_1) \vee ((q \wedge \neg p \wedge \neg p_1) \vee (q \wedge \neg p \wedge p_1)) \vee (p_1 \wedge \neg p) \\
 &= ((q \wedge \neg p_1) \vee (q \wedge \neg p \wedge \neg p_1)) \vee ((q \wedge \neg p \wedge p_1) \vee (p_1 \wedge \neg p))
 \end{aligned}$$

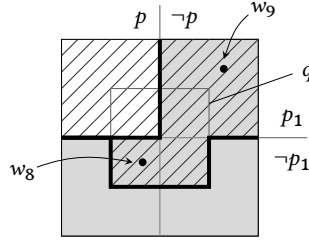


Figure 6: Exp''

Then, the prejacent p is contingent in the intersection of the new expectation set Exp'' (grey hatched area in Figure 6). Either $\neg p$ -worlds (w_9 in Figure 6), or p -worlds (w_8 in Figure 6), are possible worlds with respect to this new expectation set. Thus, q is a good answer.

At this point, we have to distinguish two cases, the case where the new piece of information is deemed to be false, and the case where it is deemed to be true. Let's start by the case where the speaker does not accept the truth of answer q . The new expectation set is $\text{Exp}''' = \{\neg q \rightarrow p_1, p_1 \rightarrow \neg p, \neg q\}$ and the possible worlds are $\bigcap \text{Exp}''' = \neg p \wedge p_1 \wedge \neg q$ (cf. the grey hatched area in Figure 7). The proposition p remains non-contingent and false in the expectation set Exp''' , i.e. q is not a good answer.

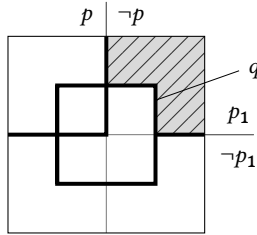


Figure 7: Exp'''

Now assume the speaker accepts the truth of answer q . The new expectation set is $\text{Exp}'''' = \{\neg q \rightarrow p_1, p_1 \rightarrow \neg p, q\}$ and the possible worlds are $\bigcap \text{Exp}'''' = (q \wedge \neg p_1) \vee (q \wedge \neg p)$ (the grey hatched area in Figure 8). The proposition p becomes contingent in the expectation set Exp'''' , and q is a good answer.

Summing up, in this section, we have seen that different statuses of the new information with respect to the expectation set, i.e. its information contribution,

$$= (q \wedge \neg p_1) \vee (p_1 \wedge \neg p)$$

for $q \wedge \neg p \wedge \neg p_1 \subseteq q \wedge \neg p_1$ and $q \wedge \neg p \wedge p_1 \subseteq p_1 \wedge \neg p$.

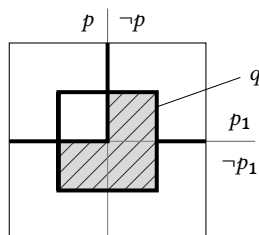


Figure 8: Exp^{'''}

lead to different modifications of the expectation set. New information that is not an issue leads the speaker to refine her default judgements, whereas new information that is an issue invites the speaker to negate some of her expectations. When this computation fails, the new information is not able to align the view-points of the interlocutors about preconditions to admitting the prejacent into the common ground.

5 Concluding remarks

This paper has explored further the hypothesis put forth by [Fleury & Tovenà \(2018\)](#) that questions with *comment* in their reason reading primarily verbalise the speaker’s attempt to bounce back from an expectation disconfirmation, and convey her request for information that would count towards admitting the prejacent in the common ground. A host of empirical observations motivate the adoption of a syntactic analysis for reason-*comment* as merged high, a position from which it can semantically operate on the speaker’s expectations.¹²

In these questions, the sentence radical does not provide the alternatives corresponding to the congruent answers *per se*. The information brought by a congruent answer allows the speaker to modify her expectations so as to undo the non-contingency of the prejacent. In our analysis, the minimal collection of relevant expectations provides a thematic way to select worlds as a function of the expectations they are consistent with. The net result is that the expectations work like the modal base found in a kratzerian approach to modality, where the worlds that are faithful to the collection of propositions called modal base (epistemic, circumstantial) are selected and form a domain the modal quantifies over. Note, however, that the partition of the worlds induced by the expectation set is not meant to represent the answers of the question, but to distinguish different

¹²An analysis in terms of syntactic operator and no variable binding has been proposed for *why* by [Stepanov & Tsai \(2008\)](#). More work is needed to assess if it suits reason-*comment*.

kinds of modifications of the expectation set brought about by different answers. An open research question is to explore whether our solution can restore the possibility of an analysis of reason-*comment* questions in terms of a quantifier binding a variable.

The speaker's expectations are characterised as a minimal set of propositions that makes the prejacent false and non-contingent. Expectations partition the possible worlds in such a way that the prejacent is not an issue with respect to this partition. Congruent answers, which may or may not be issues, enable the speaker to recompute the partition by negating or refining her expectations.

Finally, our analysis allows us to account for a difference in the reason reading between *comment* and *why* so far unnoticed, to the best of our knowledge. Suppose a context where Paul and his friends know he has to stop drinking. He doesn't, and his friends discuss about the situation. The data in (24) show that *comment* and *why* differ with respect to their compatibility with an answer that provides an explanation with no causal relation to the event.

- (24) a. Q: Comment Paul continue à boire ? A: Il continue bien à fumer.
 how Paul continues to drink he continues well to smoke
 How come Paul continues to drink? Well, he continues to smoke too.
- b. Q: Pourquoi Paul continue à boire ? A: #Il continue bien à fumer.
 why Paul continues to drink he continues well to smoke
 Why does Paul continue to drink? Well, he continues to smoke too.

Our device authorises the answer provided in (24a) to resolve the conflict. In this example, the answer resolves the conflict in virtue of the general principle that if one does not follow a recommendation in one area, it is possible/probable/not improbable that one will not follow a recommendation in a similar area. On the other hand, such an answer can hardly be regarded as a cause of the prejacent or even a reason corresponding to the similar question with *pourquoi* presented in (24b).

References

- Collins, Chris. 1991. Why and how come. *MIT Working Papers in Linguistics* 15. 31–45.
- Crespo, Inés, Hadil Karawani & Frank Veltman. 2018. Expressing expectations. In Derek Ball & Brian Rabern (eds.), *The Science of Meaning: Essays on the Metatheory of Natural Language Semantics*, 253–275. Oxford University Press.

- Cysouw, Michael. 2004. *Interrogative words: an exercise in lexical typology*. talk at 'Bantu grammar: description and theory', ZAS Berlin.
- Desmets, Marianne & Antoine Gautier. 2009. 'Comment n'y ai-je pas songé plus tôt?', questions rhétoriques en *comment*. *Travaux de linguistique* 58(1). 107–125.
- Fleury, Damien & Lucia M. Tovenà. 2018. Reason questions with *comment* are expressions of an attributional search. In *Proceedings of 22nd SemDial (AixDial)*, 112–121.
- Groenendijk, Jeroen & Martin Stokhof. 1984. *Studies on the semantics of questions and the pragmatics of answers*. University of Amsterdam. (Doctoral dissertation).
- Heider, Fritz. 1958. *The psychology of interpersonal relations*. New York: John Wiley & Sons.
- Ko, Heejeong. 2005. Syntax of *why-in-situ*: merge into [Spec,CP] in the overt syntax. *Natural language and linguistic theory* 23(4). 867–916.
- Krifka, Manfred. 2008. Basic notions of information structure. *Acta Linguistica Hungarica* 55(3–4). 243–276.
- Lewis, David. 1988. Relevant implication. *Theoria* 54(3). 161–174.
- Ochi, Masao. 2004. *How come* and other adjunct *wh*-phrases: a cross-linguistic perspective. *Language and Linguistics* 5(1). 29–57.
- Rizzi, Luigi. 1990. *Relativized minimality*. MIT Press.
- Rizzi, Luigi. 2001. On the position 'interrogative' in the left periphery of the clause. In Guglielmo Cinque & Giampaolo Salvi (eds.), *Current studies in Italian syntax: essays offered to Lorenzo Renzi*, 287–296. Amsterdam: Elsevier.
- Rullmann, Hotze. 1995. *Maximality in the semantics of wh-constructions*. University of Massachusetts, Amherst USA. (Doctoral dissertation).
- Sadock, Jerrold M. 1971. *Queclaratives*. In *Papers from the seventh regional meeting of the Chicago linguistic society*, 223–232.
- Shlonsky, Uri & Gabriela Soare. 2011. Where's 'Why'? *Linguistic Inquiry* 42(4). 651–669.
- Stepanov, Arthur & Wei-Tien Dylan Tsai. 2008. Cartography and licensing of *wh*-adjuncts: a cross-linguistic perspective. *Natural language and linguistic theory* 26(3). 589–638.
- Tsai, Wei-Tien Dylan. 2008. Left periphery and *how-why* alternations. *Journal of East Asian Linguistics* 17. 83–115.
- Veltman, Frank. 1996. Defaults in update semantics. *Journal of philosophical logic* 25(3). 221–261.
- Von Stechow, Kai & Anthony S. Gillies. 2010. Must... stay... strong! *Natural language semantics* 18(4). 351–383.

von Stechow, Arnim. 1991. Focusing and backgrounding operators. In Werner Abraham (ed.), *Discourse particles*, 37–84. Amsterdam: Benjamins.